Order-Sensitive Secular Determinants for an Inhomogeneous Coined Quantum Walk

Hénon Route-A Working Series, C143

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Abstract

We construct a spatially inhomogeneous two-state coined quantum walk on a five-cycle. Rational reflection coins and a flip-flop shift give a source-derived real unitary $U = SC$ with involutive antiunitary reversal $\Theta = CK$. Its trace expansion retains signed primitive path amplitudes. Two coin arrangements with the same local population but different spatial order have distinct degree-ten secular polynomials; their difference factors exactly. Population averaging both erases this distinction and destroys unitarity. The result reaches a unitary Route-A subgate but makes no arithmetic or target spectral identification.

1 Frozen coined dynamics

On $\mathcal{H} = \mathbb{C}^5 \otimes \mathbb{C}^2$, order the basis as $(0, +), (0, -), \ldots, (4, +), (4, -)$. Freeze the real reflection coins

$$C_0 = \frac{1}{5}\begin{pmatrix} 3 & 4 \\ 4 & -3 \end{pmatrix}, \qquad C_1 = \frac{1}{13}\begin{pmatrix} 5 & 12 \\ 12 & -5 \end{pmatrix}.$$

For a spatial word $w = w_0 \cdots w_4$, let $C_w = \bigoplus_x C_{w_x}$. The flip-flop shift is

$$S|x, +\rangle = |x + 1, -\rangle, \qquad S|x, -\rangle = |x - 1, +\rangle,$$

with positions modulo five. One coin-then-shift step is $U_w = SC_w$. This convention, clock, basis, and the determinant $D_w(z) = \det(I_{10} - zU_w)$ are fixed before comparison.

2 Unitary and antiunitary structure

Theorem 1. *For every w, U_w is real orthogonal. With K denoting coefficientwise conjugation, the antiunitary $\Theta_w = C_w K$ obeys*

$$\Theta_w^2 = I, \qquad \Theta_w U_w \Theta_w^{-1} = U_w^{-1}.$$

Moreover $P_w = |U_w|^2$ is doubly stochastic and has the same one-step support and clock.

Proof. Each displayed coin is real symmetric and squares to I; the real symmetric permutation S also satisfies $S^2 = I$. Therefore $U_w^* U_w = C_w S S C_w = I$. Since K fixes both real matrices,

$$(C_w K)^2 = C_w^2 = I, qquad (C_w K)(SC_w)(KC_w) = C_w S = U_w^{-1}.$$

The row and column sums of squared moduli of a unitary are one. ☐

This antiunitary is source-derived from the same local coin that defines the evolution; it is not added after examining the determinant.

3 Signed primitive path ownership

A directed state path carries the product of the corresponding entries of U_w, including every sign. For a primitive cycle p, write its step clock as ℓ_p and signed amplitude as A_p.

Proposition 1. *For $|z| < 5/7$,*

$$D_w(z) = \prod_{[p]} (1 - A_p z^{\ell_p}), \tag{1}$$

where $[p]$ runs over primitive directed state cycles. The left side gives the global polynomial continuation; no raw-product convergence outside the displayed disk is asserted.

Proof. Expanding $-\log D_w(z) = \sum_{n\geq 1} \operatorname{Tr}(U_w^n) z^n/n$ enumerates rooted closed paths with signed amplitudes. Primitive-root decomposition and cyclic invariance of an amplitude product give (1). Every column of $|U_w|$ has sum at most $7/5$, so the total absolute rooted-path mass is bounded by $10(7/5)^n$. This proves absolute convergence on the stated disk and justifies regrouping without discarding cancellations. □

4 Equal populations, different determinants

Compare

$$w = 00011, qquad v = 00101.$$

Both have population $(N_0, N_1) = (3, 2)$, but they are not related by a cyclic rotation or a reflection followed by rotation. Exact elimination gives

$$D_w(z) = 1 + \frac{5617}{4225} z^2 + \frac{6798}{4225} z^4 - \frac{18432}{21125} z^5 + \frac{6798}{4225} z^6 + \frac{5617}{4225} z^8 + z^{10},$$
$$D_v(z) = 1 + \frac{417}{325} z^2 + \frac{538}{325} z^4 - \frac{18432}{21125} z^5 + \frac{538}{325} z^6 + \frac{417}{325} z^8 + z^{10}.$$

Consequently

$$D_w - D_v = \frac{196}{4225} z^2 (z-1)^2 (z+1)^2 (z^2+1) \neq 0. \tag{2}$$

Spatial coin order is therefore visible to the exact determinant.

Both polynomials are palindromic. Indeed, S is a product of five transpositions and C_w of five reflections, so $\det U_w = 1$; real orthogonality then gives $D_w(z) = z^{10} D_w(1/z)$. This finite source symmetry is not a target functional equation.

5 Why population averaging is invalid

The seemingly natural averaged coin is

$$\overline{C} = \frac{3C_0 + 2C_1}{5} = \frac{1}{325} \begin{pmatrix} 167 & 276 \\ 276 & -167 \end{pmatrix}.$$

It satisfies

$$\overline{C}^{,T} \overline{C} - I = -\frac{24}{1625} I, qquad \det \overline{C} = -\frac{1601}{1625}.$$

Thus averaging destroys unitarity as well as order. It is a negative control, not an alternative representation of either walk.

6 Exact receipt and scope

For each arrangement the evidence reconstructs 12 traces and every signed rooted path through clock ten; at clock ten there are 1,270 rooted closed paths and 125 primitive cycles. The producer-independent checker makes 62 assertions, a separate SymPy route makes 39 exact checks, replay is byte-identical, and 29 repaired-hash plus one stale-hash mutations are rejected.

The strict verdict is

$$(\mathrm{A1\_WEAK}, \mathrm{A2\_FAIL}, \mathrm{A3\_FAIL}, \mathrm{A4\_UNITARY\_OR\_SCATTERING\_CANDIDATE}).$$

There is no target divisor, missing/extra-zero census, target counting law, prime correspondence, arithmetic local factor, Euler factor, root number, automorphy claim, or self-adjoint Hilbert–Pólya operator. The dimension is fixed and no growing-level limit is claimed. Route B remains unauthorized under `NO_BAD_EULER_OR_ROOT_NUMBER`.

7 Conclusion

C143 shows that one source-derived discrete unitary can retain chronological spatial order, signed orbit amplitudes, and exact reversal simultaneously. Connecting this finite walk to any external target remains open.